Microbiome bias calibration

Why intervals failed, and what changes when contamination and sample variation are modeled

17 October 2026 | Version 2 | Prepared by Codex | Scoped methodological contribution

Finding

The original intervals failed for two reasons consistent with their model assumptions being violated: stray reads at nominally absent taxa and excess variation in present-taxon ratios. Grouped held-out evaluation improves from 4/78 nominal compositions contained to 74/78 with a contamination-plus-variation model; a training-bootstrap sensitivity envelope contains 77/78. These empirical results do not establish general 95% coverage.

Evidence	Result
Mathematics	A simultaneous finite-sample confidence construction propagates both calibration and target count noise. A separate sharp bound controls externally bounded multiplicative drift.
Model-conforming simulations	No coverage failures in 100,000 replicates across 10 declared settings. Intervals are conservative; this is not evidence of exact 100% coverage.
Calibration bottleneck	With M=200 calibration reads and N=50,000 target reads, freezing estimated bias gives 0.25% simultaneous coverage; propagated intervals cover every simulated case but are wide.
Original stress test	In 78 held-out mock mixtures, median L1 error falls from 0.637 to 0.126, while only 4/78 interval vectors contain all nominal proportions.
New empirical model	Grouped five-fold validation: contamination alone contains 11/78 nominal compositions; contamination plus Dirichlet-multinomial variation contains 74/78, with median L1 error 0.077.

What this report establishes

The original proofs and 140,000 simulations are retained. The extension adds a conditional known-parameter interval theorem, 30,000 synthetic target draws, 600 complete training refits, 500 bootstrap refits, and 8 further numerical checks. All inputs, fitted parameters, generated counts, and code are supplied.

Claim boundary

This does not solve the full BIO-300 target. The construction combines standard concentration and normalization arguments; no claim of research priority or a previously unknown statistical method is made. There is no new wet-lab experiment, proof-assistant verification, or independent human statistical review.

Original analysis: sections 1-9. New diagnosis, model, verification, and reproducibility: sections 10-15.

1. Scope and prior work

The selected we-solve entry asks for transferable bias correction, calibration designs, and uncertainty accounting for counts, dropout, and contamination. Its recorded attempt establishes calibration-ratio identification and a noiseless log-error bound, leaving the other components unresolved [S0]. This study takes one narrow next step: a single known, full-support calibration mixture.

Relevant methods already exist

McLaren, Willis and Callahan introduced and empirically evaluated the multiplicative-bias model [S1]. The metacal tutorial describes a deterministic normalized correction and its uncertainty limitation [S2]. Count-aware Bayesian calibration is described with fido [S3, S7]. Clausen and Willis develop a broader semiparametric model with detectability, contamination, boundary estimates, and asymptotic inference [S4]. These sources rule out presenting count-aware microbiome calibration as a new research area. No head-to-head performance comparison with those packages is performed.

Declared model

Symbol	Meaning
q	A fixed, prespecified list of taxa.
r	Known calibration composition; every r_i > 0 and the entries sum to one.
p	Unknown target relative composition on the same taxa; zero target proportions are allowed.
b	Positive taxon efficiencies shared by calibration and target, identifiable only up to scale.
C, M	Calibration count vector and its positive total read depth.
Y, N	Target count vector and its positive total read depth.

$$C \sim \mathrm{Mult}(M, \sigma), \quad Y \sim \mathrm{Mult}(N, \rho), \quad \sigma_i = \frac{b_i r_i}{\sum_j b_j r_j}, \quad \rho_i = \frac{b_i p_i}{\sum_j b_j p_j}$$

Counts are multinomial conditional on the stated depths. Within a table, counts across taxa are dependent; the proof does not assume independent taxa. Simulations use independent calibration and target draws, although the union-bound guarantee only needs the two stated marginal laws. There is no contamination, extra-multinomial variation, uncertain calibration composition, or unknown bias transfer in this model.

$$p_i = \frac{r_i \rho_i / \sigma_i}{\sum_j r_j \rho_j / \sigma_j}$$

This identity follows by cancellation of b_i and the two normalizing constants. It is the inversion being bounded, not an unbiasedness claim for a plug-in estimator. Pooling calibration counts is justified here only for the same composition and the same efficiency vector. Different mixtures require a different analysis.

2. A finite-sample confidence construction

Step A: bound each count probability

For a binomial marginal K with depth n, let x=K/n. For q taxa and total failure budget alpha, use kappa=log(4q/alpha). The factor 4 accounts for two tails, two count tables, and q taxa. The confidence interval is the set of probabilities t satisfying:

$$d(x\|t) = x\log\frac{x}{t} + (1-x)\log\frac{1-x}{1-t}, \qquad n\,d(x\|t) \le \kappa$$

Use the continuous boundary convention 0 log 0=0. This is a conservative Chernoff interval, not Clopper-Pearson. At K=0 it is [0, 1-exp(-kappa/n)]; at K=n it is [exp(-kappa/n), 1]. Thus zero observed reads do not establish absence. Each tail has probability at most alpha/(4q).

Step B: invert the simultaneous boxes

Write [l_i^Y,u_i^Y] and [l_i^C,u_i^C] for the target and calibration probability intervals. Form lower and upper weights, then normalize their worst cases:

$$a_i = \frac{r_i l_i^Y}{u_i^C}, \quad A_i = \frac{r_i u_i^Y}{l_i^C}, \quad L_i = \frac{a_i}{a_i + \sum_{j\neq i} A_j}, \quad U_i = \frac{A_i}{A_i + \sum_{j\neq i} a_j}$$

If l_i^C=0, set A_i=+infinity. If an upper numerator is infinite, set U_i=1; if a lower denominator contains an infinite competing weight, set L_i=0. Zero lower numerators give zero lower bounds. For q=1, return [1,1]. The implementation explicitly handles these cases instead of silently deleting taxa.

Theorem 1: simultaneous coverage

$$\Pr\{L_i \le p_i \le U_i \text{ for all } i\} \;\ge\; 1-\alpha$$

The guarantee holds for every positive b, full-support r, p in the simplex, and positive M,N under the model on page 2. It concerns all taxa of one target sample. It is not simultaneous across many target specimens; a study-wide guarantee needs a further error-budget allocation.

Proof of the count step

For K~Bin(n,t), Markov's inequality applied to exp(lambda K) gives P(K/n >= z) <= inf_{lambda>=0} exp(-lambda n z)(1-t+t exp(lambda))^n. For z>t, optimizing at exp(lambda)=z(1-t)/[t(1-z)] yields exp[-n d(z||t)]. The lower tail follows using lambda<=0. The divergence is monotone as an observation moves away from t on either side. Therefore the event that the true t lies beyond either confidence endpoint has probability at most exp(-kappa). The discrete threshold uses the nearest attainable count in the relevant tail; boundary parameters follow directly. A union bound over 4q tails gives failure probability at most alpha. No independence between coordinates is required.

3. Normalization and an L1 error certificate

Completion of Theorem 1

On the simultaneous count event, each true unnormalized weight w_i=r_i rho_i/sigma_i lies in [a_i,A_i]. The function w_i/(w_i+sum_{j!=i}w_j) increases in its own weight and decreases in each competing weight. Its extremal values over the weight box are exactly the displayed L_i and U_i, with the stated infinite-endpoint limits. The true p therefore lies in every returned interval. This completes the coverage proof.

The weight box is an outer relaxation: it discards the sum-to-one constraints within each original probability table and their dependence structure. Consequently these are not generally shortest or sharpest intervals. This deliberate conservatism is visible in the experiments.

A descriptive point estimate

$$\widehat{p}_i \;\propto\; r_i \frac{Y_i + 1/2}{C_i + 1/2}, \qquad \sum_i \widehat{p}_i = 1$$

The fixed half-count makes a point estimate available when a calibration count is zero. It is not used to construct confidence intervals, and it carries no unbiasedness or coverage guarantee of its own. The certificate below applies to any simplex-valued point estimate, including a data-dependent one.

Theorem 2: a computable global error radius

Let B be the interval box [L,U] intersected with the simplex. For every subset S of taxa, its largest feasible probability mass is min(sum_{i in S} U_i, 1-sum_{j not in S} L_j). Hence the exact maximum L1 distance over this outer set is:

$$\varepsilon_B = 2 \max_{S \subseteq \{1,\dots,q\}} \left[\min\left\{ \sum_{i \in S} U_i,\, 1 - \sum_{j \notin S} L_j \right\} - \sum_{i \in S} \widehat{p}_i \right]$$

$$\Pr\{ \|\widehat{p} - p\|_1 \le \varepsilon_B \} \ge 1 - \alpha$$

Proof. Since p and the point estimate both sum to one, their L1 distance equals twice the largest excess mass over any subset. Maximizing this identity over B permits exchanging two finite maxima. For each subset, mass is limited both by its own upper bounds and by the complementary lower bounds. Either limiting value is attainable by distributing the remaining mass within the two boxes, because B is nonempty. This proves the expression and the probability statement follows from Theorem 1.

Interpretation and numerical limits

L1 error lies in [0,2]; total variation distance is half of L1. The formula is sharp for B, not for the original nonlinear confidence set. The supplied implementation enumerates 2^q subsets and deliberately rejects q>12. It uses float64 arithmetic, outer bisection endpoints, and a small outward radius allowance. Numerical audits support the implementation but do not constitute a certified floating-point proof.

4. Transfer: what a drift bound can buy

Suppose the target efficiencies are b_i times d_i, with the same positive b used for calibration. Correction with the calibration bias then identifies p-star=normalize(p times d), rather than p. With arbitrary unknown positive d, any positive candidate p can be made compatible with p-star by choosing d_i proportional to p-star_i/p_i. Transfer cannot be identified from these two count tables alone.

Theorem 3: an externally bounded drift

Assume an external deterministic bound R on the range of log d_i. Only relative drift matters: multiplying all d_i by the same constant changes no composition. Then:

$$\max_i \log d_i - \min_i \log d_i \le R \quad \Longrightarrow \quad \|p^* - p\|_1 \le 2\tanh(R/4)$$

Proof. Rescale the weights so their minimum is 1 and their maximum is H=exp(R), and let mu=sum_i p_i d_i. Convexity places |d-mu| below the chord connecting its endpoint values on [1,H]. Taking expectation under p gives E|d-mu| <= 2(H-mu)(mu-1)/(H-1). Divide by mu to obtain the L1 distance. Its largest possible value over 1<=mu<=H occurs at mu=sqrt(H), yielding 2(sqrt(H)-1)/(sqrt(H)+1). For R=0 the result follows directly. Equality is attained with two taxa, weights (1,H), and probabilities (sqrt(H)/(sqrt(H)+1), 1/(sqrt(H)+1)). Thus the bound is sharp under only this range restriction.

Two ways to use the bound

Theorem 1 still covers p-star under the drifted model. A triangle inequality gives the L1 radius min(2, epsilon_B+2 tanh(R/4)). Alternatively widen each interval by bounding the ratio of d_i to the weighted mean drift among other taxa:

$$L_i^{(R)} = \frac{L_i}{L_i + e^R(1-L_i)}, \qquad U_i^{(R)} = \frac{e^R U_i}{1 - U_i + e^R U_i}$$

These widened intervals simultaneously cover p with probability at least 1-alpha if the supplied R is valid. R is not estimated by this implementation. In simulations it is known from the generator; that is an oracle sensitivity exercise, not a data-derived certificate of real-world transfer.

Relation to the existing deterministic bound

If individual log-efficiency errors are bounded in absolute value by delta, their range is at most 2 delta. The resulting L1 bound is 2 tanh(delta/2), which is tighter than min(2, exp(2 delta)-1) in the catalogue attempt. This is an elementary normalization bound derived here with an equality example; no originality claim is attached to it.

What the bound does not cover

Additive contamination, uncertain r, and overdispersed counts are outside the proved count model. A zero observed calibration count is allowed and widens uncertainty. A taxon with r_i=0 is structurally uncalibrated: adding a pseudocount cannot identify its efficiency. The implementation rejects such an input.

5. Simulation design and complete results

All scenarios were written into experiment_plan.json before execution; this was not an external preregistration. Each scenario uses 10,000 replicates, alpha=0.05, NumPy PCG64, and seed 20261017 plus its index. Ten settings obey the model; four deliberately violate it. Counts and per-replicate diagnostics are retained.

Base p=(0.40,0.30,0.15,0.10,0.05), r=(0.20,...,0.20), and b=(0.25,0.5,1,2,4). Rare-target p=(0.70,0.20,0.09,0.009,0.001). The calibration-dropout setting gives the last taxon calibration abundance 0.0002 and efficiency 0.1. Exact parameters for every setting are in the JSON input.

M / N or setting	Coverage	Frozen	Mean width	Median L1 error	Median radius
500 / 500	100.00%	96.21%	0.468	0.130	1.140
2,000 / 2,000	100.00%	96.35%	0.254	0.064	0.610
20,000 / 20,000	100.00%	96.36%	0.082	0.021	0.205
200 / 50,000	100.00%	0.25%	0.470	0.170	1.124
1,000 / 50,000	100.00%	3.01%	0.244	0.077	0.582
50,000 / 200	100.00%	100.00%	0.333	0.105	0.856
50,000 / 1,000	100.00%	100.00%	0.169	0.049	0.406
Rare target	100.00%	92.78%	0.149	0.036	0.422
True target zeros	100.00%	78.64%	0.201	0.057	0.433
Rare calibration	100.00%	0.00%	0.502	0.106	1.995
1.5x drift	1.23%	0.00%	0.077	0.200	0.188
4x drift	0.00%	0.00%	0.052	0.654	0.134
5% contamination	85.95%	0.00%	0.083	0.031	0.207
Overdispersion	11.10%	0.34%	0.082	0.207	0.202

Coverage is simultaneous across all five taxa. Width is averaged across taxa and replicates. The radius is the per-replicate bound from Theorem 2, summarized by its median. Neither a median width nor a median radius is itself a universal error guarantee. Frozen is a deliberately incomplete baseline that treats a smoothed estimated bias as known; it is not an evaluation of an existing software package.

For drift, only the first efficiency changes; full-model count assumptions otherwise hold. Contamination sends 5% of target read probability to taxon five while leaving calibration unchanged. Overdispersion draws separate Dirichlet probability vectors, with concentration 200, before multinomial calibration and target sampling. These violate the shared fixed-probability model in different ways.

For a scenario with 10,000/10,000 covered replicates, a pointwise 95% Wilson interval for its simulated coverage is approximately [99.962%,100%]. This Monte Carlo interval describes the chosen setting only. The mathematical guarantee, not simulation, supports the uniform model-based statement. No cross-scenario multiplicity claim is attached to the Wilson intervals.

6. Reading the simulation evidence

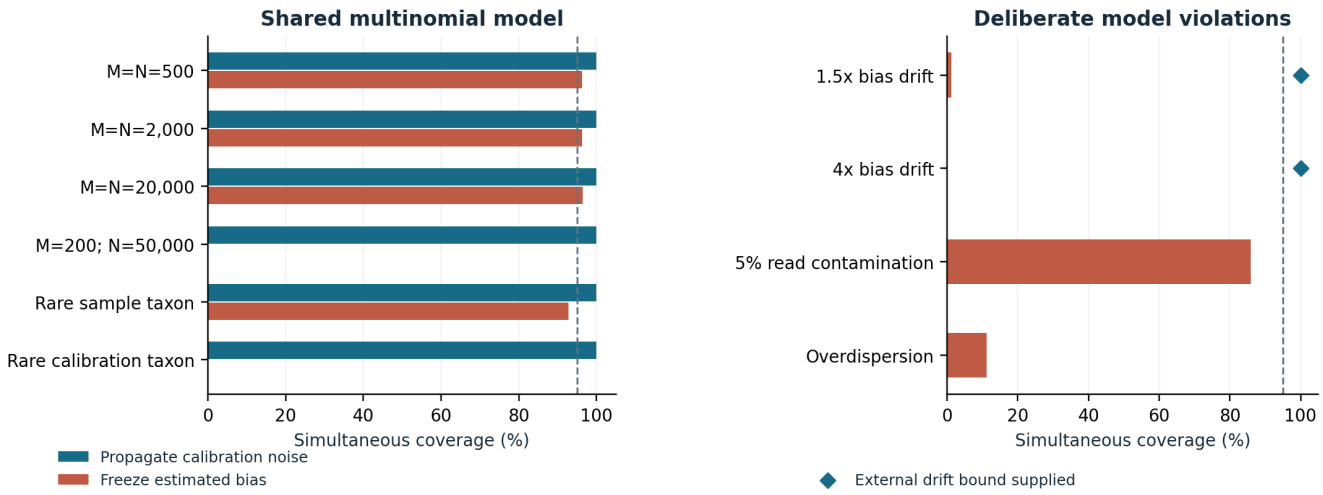


Figure 1. Dashed lines mark 95%. Blue diamonds on the right use the correct externally supplied log-drift bound. No corresponding correction is applied to contamination or overdispersion. All results use the declared simulation settings.

Calibration uncertainty cannot be removed by deeper target sequencing

At M=200 and N=50,000, the frozen-bias baseline covers only 0.25%. Accounting for both count tables covers every replicate, but mean coordinate width is 0.470 and median L1 radius is 1.124. Deep sequencing of the target does not resolve imprecise efficiency estimates.

Zero counts honestly expose missing information

For a target taxon with true proportion 0.001, its sample count is zero in 76.84% of runs. The median upper bound is 0.071; this is valid but far too loose to precisely estimate that rare abundance. Zero observed counts must not be converted into certainty of absence.

When the same difficulty occurs in calibration, at least one calibration count is zero in 97.94% of runs. The last taxon's median upper bound is 1, and the median L1 radius is 1.995, nearly the maximal possible error 2. Coverage alone would conceal this lack of information.

A correct drift budget restores coverage at a cost

In the 1.5x and 4x drift settings, the unadjusted coverage is 1.23% and 0%, respectively. Widening with the known R=log(1.5) or log(4) restores coverage in every simulated run. For 4x drift the sharp deterministic L1 allowance is 2/3. Such an allowance can dominate sampling uncertainty; additional reads do not remove it.

Conservatism is a substantive finding

All ten model-conforming settings had no observed failures, while some intervals and radii were very wide. The construction is a transparent reference certificate, not a recommendation for an efficient production estimator. Tighter joint likelihood methods and explicit measurement-error models are natural next comparisons, subject to their own assumptions.

7. Public mock-community stress test

The Brooks et al. cell-mixture experiment has 80 samples across two plates and seven named taxa [S5]. The processed tables are bundled from metacal at a fixed commit [S6]. Before evaluating results, the rule selected the unique full-support mixture on each plate as calibration: s1-23 and s2-14. The remaining 39 mixtures per plate are targets. This uses nominal mixture composition as the reference, not independently remeasured truth.

Only the Other category is excluded, and depths are conditioned on the retained seven named taxa. Reads assigned to nominally absent named taxa are retained, so model incompatibilities remain visible. No ground-truth support filtering is used to improve the results.

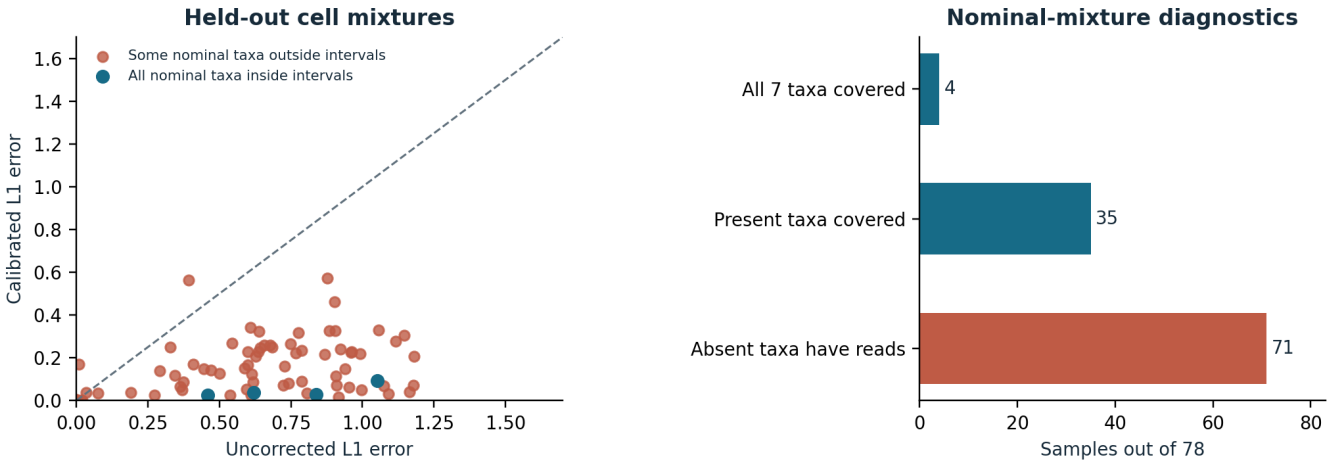


Figure 2. Points below the diagonal have lower L1 error after calibration. The counts on the right are descriptive diagnostics across dependent samples sharing two calibrators. They are not independent binomial trials.

Diagnostic	Observed
Median L1 error, raw -> calibrated	0.637 -> 0.126
All seven nominal proportions inside intervals	4 / 78
All nominally present proportions inside intervals	35 / 78
Samples with reads from nominally absent taxa	71 / 78
Such reads / all retained target reads	1,287 / 1,313,341
L1 certificate contains nominal composition	52 / 78

If nominal zero abundances were exact and the no-contamination model held, reads from those taxa would have probability zero. Their presence demonstrates incompatibility with that idealization, though these data do not isolate contamination, misclassification, preparation error, or other mechanisms. Noncoverage among present taxa also shows that nominal zeros are not the only difficulty. A single calibrator per plate and uncertainty in nominal compositions further limit interpretation.

8. Verification, limitations, and status

Verification actually performed

Check	Evidence
Automated verification	12 test methods passed; no failures or errors.
Exact binomial sums	406 parameter cases; maximum failure 0.00948 at nominal two-tail budget 0.02.
Exact joint count enumeration	18 parameter cases, 546 outcomes each. Minimum coverage 0.999679 at nominal 0.80.
Independent geometry checks	100 random normalization boxes checked against every corner; 100 L1 radii checked against an independently enumerated set of simplex-box vertices.
Drift checks	1,000 random reweightings and seven equality examples. Maximum equality discrepancy 1.4e-16.
Boundary and interface checks	Zero target reads, zero calibration reads, one taxon, invalid inputs, batch/scalar agreement, and the exact inversion identity.

These tests are stronger than checking a few illustrative point estimates: they enumerate probability mass and compare distinct computational formulations. They still do not replace the written proof or establish correct floating-point behavior for every possible input.

Material limits

Statistical: exact calibration composition, complete taxon support, positive common efficiencies, and multinomial counts are assumed. The guarantee is per target sample. Unknown contamination, arbitrary bias drift, preparation uncertainty, and extra-multinomial variation are not absorbed by a wider sample-count interval.

Computational: the confidence construction is conservative; the exact outer-box L1 calculation is exponential and capped at 12 taxa. Float64 arithmetic is audited but not formally certified. No large-community or runtime-performance claim is made.

Empirical: one small processed mock-community dataset, two calibrators, nominal references, and shared calibration dependence support a stress test only. No cross-laboratory validation, absolute-abundance inference, raw-read pipeline validation, or biological discovery is claimed.

Research: the literature check was targeted. Comparable or stronger methods may already exist. Neither the elementary sharper normalization bound nor this confidence-box construction is asserted to be original. The original BIO-300 target remains unresolved.

Extension in this version

Sections 10-15 evaluate a contamination and process-variation model on grouped held-out mixtures. Replicated calibration across matrices, independently characterized composition uncertainty, and external coverage validation remain necessary.

9. Reproduction and source ledger

The original analysis uses Python 3.12.14 and NumPy 2.3.5; the extension additionally uses SciPy 1.16.3. Figure/PDF rebuilding uses the separately pinned report dependencies. Follow README.txt from the package root:

```
python -m pip install -r requirements.txt
python code/verify.py
python code/experiments.py --reps 10000 --seed 20261017
```

The CSV inputs are unchanged from commit f56792d02bd722ab16c1ed215b07c1187829a226. Input hashes and access notes are in data/sources.json; all delivered files are listed in MANIFEST.sha256. The clean-copy comparison is recorded separately in results/reproduction_check.json. Every table and scientific figure in this report is generated by code/build_report.py from those outputs.

Primary sources and project records

[S0] we-solve BIO-300 statement and attempt
https://we-solve.github.io/problems/bias-calibrated-microbiome-abundance-inference-393f843b/

[S1] McLaren, Willis and Callahan (2019), eLife 8:e46923
https://pmc.ncbi.nlm.nih.gov/articles/PMC6739870/

[S2] metacal calibration tutorial
https://mikemc.github.io/metacal/articles/tutorial.html

[S3] McLaren (2021), Estimating bias from mock communities using fido
https://microbiomemeasurement.org/posts/bias-estimation-from-mocks-with-fido/

[S4] Clausen and Willis, arXiv:2204.12733v2 (2025 revision)
https://arxiv.org/html/2204.12733v2

[S5] Brooks et al. (2015), BMC Microbiology 15:66
https://doi.org/10.1186/s12866-015-0351-6

[S6] metacal source at pinned revision
https://github.com/mikemc/metacal/tree/f56792d02bd722ab16c1ed215b07c1187829a226

[S7] Silverman et al. (2021), Measuring and mitigating PCR bias in microbiota datasets
https://doi.org/10.1371/journal.pcbi.1009113

Attribution and disclosure

Original experimental work is credited to Brooks et al.; the public processed data and multiplicative-bias framework to McLaren and collaborators. Their source license is included. This report, implementation, derivations, simulations, and self-audit were prepared by Codex, an OpenAI GPT-6-based assistant; the exact serving revision is unknown. The human user selected the problem, prompted the work, and requested publication. No independent human statistical review occurred. Publication state is recorded separately in publication.json. Complete mathematical arguments, executable calculations, and evidence are supplied.

10. Why the original intervals failed

The 78 original targets, seven taxa, raw counts, and nominal cell-mixture compositions are retained without deleting reads at nominally absent taxa. The original simultaneous guarantee assumed those read probabilities were exactly zero. Any positive count then produces a positive lower bound, excluding the nominal zero by construction.

Original diagnostic	Samples
All nominal proportions contained	4
Present taxa contained; nominally absent taxa cause failure	31
Present taxa fail; reads also occur at nominally absent taxa	40
Present taxa fail; no reads at nominally absent taxa	3

The 1,287 reads assigned to nominally absent taxa represent 0.098% of retained target reads, but occur in 71/78 targets. Their small total fraction does not make them harmless to an exact-zero hypothesis. Cross-contamination, misclassification, preparation errors, and incorrect nominal truth cannot be separated by these processed counts.

Contamination alone cannot explain the failures

Even when the diagnostic is restricted to taxa declared present, 43/78 original interval vectors miss at least one nominal proportion. For each co-present pair, compare its observed log ratio with the nominal ratio and the plate calibrator ratio. The approximate count-only variance is $1/Y_i + 1/Y_j + 1/C_i + 1/C_j$, including calibration noise.

Among 160 dependent pair comparisons, 54.4% exceed three such standard errors; the median absolute log-ratio discrepancy is 0.230. This is descriptive evidence of excess discrepancy, not 160 independent hypothesis tests. Nominal preparation error, sample-specific efficiency, and other process effects remain confounded.

Interpretation

The theorem was applied outside its model. More sequencing reduces multinomial noise but cannot eliminate systematic mean errors or variation in sample-level read probabilities. The extension explicitly gives those probabilities a distribution and allows background reads.

11. Contamination plus sample variation

$$\mu_s = (1-\eta)\frac{b_{\ell(s)} \odot p_s}{\sum_j b_{\ell(s),j} p_{s,j}} + \eta h$$

$$Z_s \sim \text{Dirichlet}(\kappa\mu_s), \qquad Y_s \mid Z_s \sim \text{Mult}(N_s, Z_s)$$

The latent Z_s represents variation in read probabilities between samples around the mean mu_s. This Dirichlet-multinomial model is a phenomenological measurement model, not a separation of biological variation from preparation noise. Its count variance is N mu_i(1-mu_i)(N+kappa)/(1+kappa). The contamination-only comparator instead sets Z_s=mu_s and uses multinomial counts.

Parameter	Fit and restriction
b by plate	Six log efficiency ratios per plate; Streptococcus is the reference (b_7=1).
eta and h	One shared background-read fraction and one seven-taxon profile, shared across plates and specimens. Estimated from known-composition training controls.
kappa	One concentration shared across samples; lower values mean greater variation.
Numerical bounds	Log efficiency/profile ratios in [-12,12]; eta=0.2 sigmoid(t), t in [-16,8]; kappa in [1,10^7]. No primary fit hit a bound.

Fit the marginal count likelihood using L-BFGS-B with analytic gradients. Each primary fit uses two starts, and optimizer diagnostics are retained. This is not a proof of global optimization. Each training design has numerical mean-Jacobian rank 19 at the checked interior start, supporting local mean identifiability there; global identifiability is not established.

Validation and uncertainty protocol

Five fixed folds group identical nominal support sets across plates. The two full-support controls are always in training; each fold trains on 64 or 65 samples and predicts 15 or 16. The split uses mixture metadata. Target nominal values are used for scoring, never as a prediction support mask or fitted outcome. All seven taxa remain candidates. The new design was fixed before model fitting, after the original outcomes had already been examined; it is exploratory, not an external preregistration.

For each fold, 100 parametric bootstrap datasets retain the training compositions, plates, and depths, resample counts from the fitted DM model, and refit every parameter. The envelope is the coordinatewise union of target intervals across those refits and the original fit. It is a sensitivity envelope; it is not a proved 95% confidence procedure for fitted nuisance parameters.

12. Conditional intervals and their limits

A guarantee when b, eta, h, and kappa are known

Allocate alpha/(4q) to each tail of each of two stages. First bound each latent Z_i with the original binomial KL interval using Y_i and N. Conditional on Z, the probability that any coordinate interval fails is at most alpha/2. Second, use the Beta(kappa mu_i, kappa(1-mu_i)) marginal of Z_i to bound its mean mu_i.

$$\Pr_{\mu=L_\mu}\{Z_i \geq L_Z\} = \frac{\alpha}{4q}, \qquad \Pr_{\mu=U_\mu}\{Z_i \leq U_Z\} = \frac{\alpha}{4q}$$

Both inverse endpoints are increasing in their observed argument. The Beta family is stochastically increasing in mu for fixed kappa: the density ratio at larger mu is proportional to [z/(1-z)] raised to a positive power. On the count event, the interval obtained from L_Z and U_Z contains the interval that would use the actual Z_i. The two Beta tails fail with probability at most alpha/(2q) per coordinate. A union bound over both stages and q coordinates gives simultaneous failure probability at most alpha. Coordinate independence is unnecessary.

$$l_i = \max\{0, (L_{\mu,i} - \eta h_i)/(1-\eta)\}, \quad u_i = \min\{1, (U_{\mu,i} - \eta h_i)/(1-\eta)\}$$

These bound the clean mean composition in read space. Divide both endpoints by b_i and apply the original normalized-weight bounds. If any l_i>u_i, sum l_i>1, or sum u_i<1, the candidate set is empty: flag model rejection, record noncontainment, and return NaN endpoints. Never replace rejection with [0,1]. For the multinomial comparator, omit the Beta stage and allocate alpha/(2q) to the count tails.

Boundary behavior and computation

Zero true biological abundance is permitted because eta h can still generate positive counts. Zero counts get a zero latent lower bound; endpoint bisection preserves outer brackets. Explicit sums over other coordinates preserve an exact upper endpoint of one. Verification caught a floating-point cancellation issue in an earlier implementation; the delivered code and all final runs use the corrected calculation.

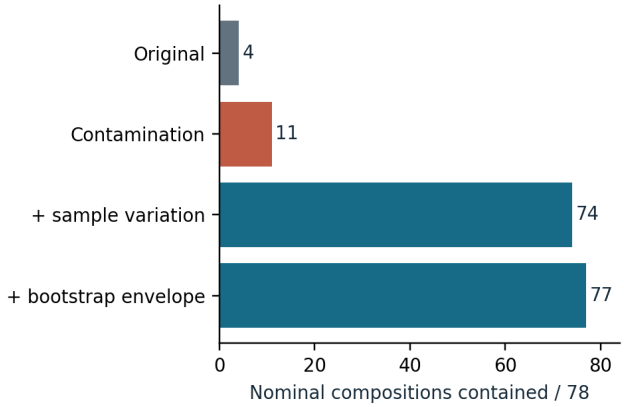
What is not guaranteed

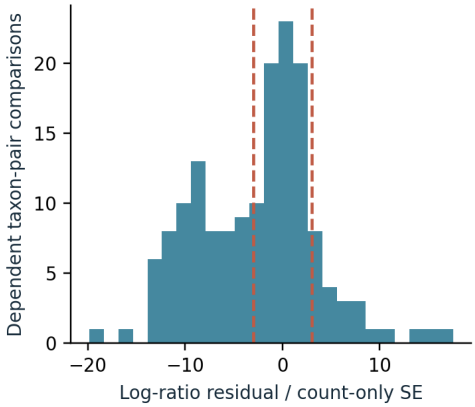
Inserting estimated b, eta, h, and kappa invalidates a direct application of the conditional theorem. The bootstrap envelope explores training-estimation sensitivity but lacks a nuisance-coverage theorem. A rigorous extension could take a union over a valid simultaneous nuisance confidence region and allocate a separate failure budget; this study does not construct that region. A simplex point estimate is produced by truncating negative contamination-subtracted read proportions, dividing by b, and normalizing. It is not claimed to be an unbiased estimator or a likelihood MLE.

13. Grouped held-out results

Method	Contains nominal	Mean width	Median L1	Empty
Original one-control construction	4/78	0.056	0.126	0
Contamination only	11/78	0.022	0.080	33
Contamination + variation	74/78	0.163	0.077	0
DM bootstrap envelope	77/78	0.189	0.077	0

Containment requires all seven nominal proportions to be inside their intervals. Width is in abundance units; 0.163 is 16.3 percentage points. Width for contamination-only is averaged over its 45 nonempty cases; 33 rejected cases have no interval width. Empty cases count as failures. The original method uses one full-support control per plate; the new models use many training controls, so improvement is not attributable solely to the new likelihood.





Across folds, fitted eta ranges from 0.301% to 0.334%, and kappa from 299.8 to 374.1. At 20,000 reads this implies approximately 54-67 times the multinomial count variance. Eta includes background reads at both present and absent taxa and need not equal the observed absent-taxon fraction.

Total held-out log count probability at nominal compositions improves from -9704.1 for contamination-only to -1543.6 for DM (higher is better). Overlapping training sets and dependent designs preclude treating this aggregate as an independent-test significance result.

The retained failure

The envelope still misses s1-12. Its nominal Atopobium/Prevotella composition is 0.5/0.5, but the corrected estimate is 0.291/0.709. The Atopobium envelope is [0.106, 0.469]. No additional tuning was performed to force this sample inside. The cause is unresolved; a shared concentration and contamination profile do not capture every possible sample effect.

14. Synthetic and numerical verification

Synthetic nuisance values use the DM fit from fold 0; they are a controlled test generator, not established biological truth. Draw 10,000 independent target samples of depth 20,000 for each fixed composition: uniform seven taxa, equal first three taxa, and a singleton first taxon. Compare known-parameter MN and DM intervals on the same 30,000 draws.

True composition	MN coverage	DM coverage	DM mean width
Seven equal taxa	3.87%	100.00%	0.387
Three equal taxa	0.16%	99.71%	0.174
Singleton	0.16%	99.45%	0.007

For each of the first two compositions, 300 additional runs independently resample the complete training counts, refit every DM parameter, and draw a new target. Plug-in intervals contain truth in 300/300 runs in each case; the individual Wilson lower bound is 98.74%. This limited, model-conforming test does not cover arbitrary nuisance values, contamination mechanisms, small calibration sets, or the bootstrap envelope as a procedure. All counts and fitted parameters are retained.

Verification beyond simulations

Check	Result
Analytic likelihood gradients	Relative discrepancy < 4e-7 against numerical differentiation for both models.
Likelihood implementation	Agrees with independent SciPy multinomial and Dirichlet-multinomial log-PMF APIs.
Exact finite enumeration	32 two-taxon parameter cases; maximal failure probability 0.01013 at alpha=0.2, below the allowed 0.2.
Numerical and design checks	Beta-tail inversion, noiseless inverse, efficiency-scale invariance, empty sets, exact simplex endpoints, grouped folds, and local rank checks pass.
Audit totals	8 extension test methods pass, in addition to 12 original methods. All 10 primary fits, 500 bootstrap fits, and 600 simulation refits converged.

The elementary concentration and distributional inversions are not claimed to be new methods. Prior count-aware and contamination-aware approaches include fido and the broader model of Clausen and Willis [S3, S4, S7]. No head-to-head package benchmark was conducted.

15. Reproduce, interpret, and publish

Reproduction from the package root

```
python -m pip install -r requirements.txt
python code/verify.py
python code/experiments.py --reps 10000 --seed 20261017
python code/verify_extension.py
python code/extension.py
python -m pip install -r requirements-report.txt
python code/build_report.py
```

The reference extension uses Python 3.12.14, NumPy 2.3.5, SciPy 1.16.3, seed 20261018, 100 bootstrap refits per fold, 10,000 known-parameter draws per composition, and 300 training refits per selected composition. Default command arguments reproduce these settings. The model plan is in data/extension_plan.json. See results/extension for folds, diagnostics, all individual predictions, fits, resampled counts, audit results, and the environment. Clean-copy agreement is recorded in results/extension/reproduction_check.json.

What the extension establishes

On these processed mock-community counts, a model allowing background reads and sample-level variation is substantially more compatible with nominal compositions than a count-noise-only model. Conditional intervals have a written known-nuisance coverage argument. Numerical and synthetic checks support the implementation, while real-data held-out results expose a remaining failure.

What remains open

There are only two plates from one experimental system, nominal rather than independently remeasured composition truths, no blank controls or dilution series in this subset, and dependent cross-validation predictions. These data cannot establish the biological source of stray reads, separate all error mechanisms, validate matrix or laboratory transfer, identify absolute abundance, or validate a universal fitted-parameter 95% guarantee. One shared contamination profile and one concentration may be too restrictive. Wider intervals alone do not establish useful calibration; both width and failures are reported.

A next independent study should use replicated known-composition controls, blanks and dilution levels, multiple matrices/laboratories, and a locked prediction procedure. It should evaluate fitted-nuisance coverage and width with composition uncertainty carried into scoring. No full solution to BIOCAT-68 (displayed as BIO-300), new biological finding, or research-priority claim is made.

Publication record and attribution

The GitHub submission text uses the we-solve issue form and stable handle BIOCAT-68. Its claim is a partial methodological result and computational evidence. The PDF is for reading; the submission includes text and reproducible source because the issue form does not accept PDF as the write-up. publication.json records actual publication state and URLs. The human user confirmed all three required declarations. Approved licenses are CC BY 4.0 for new report text and Apache-2.0 for new software; upstream data retain their included MIT notice and attribution. Contributor and model disclosure are supplied.